

DNN layer shapes

hardware: 128×128 CBs,
 c CBs per tile

1 Architecture & packing

PD: density c (hardware-fixed); PO: orientation
(r, s), $r \cdot s = c$

choose (r, s): min PF

out: traffic graph (rates λ_f , windows), PF

traffic graph

2 Mapping (tile $\rightarrow$ node)

place at min CC; refine: max ES

s.t. $CC \leq CC_{\min}$, $PL \leq PL(CC_{\min})$

out: placement π , PL, CC

placement π

3 Routing & selection (run time)

odd-even-balanced routing: admissible paths $\mathcal{R}$

selection: BL (1-hop buffer) / DP (multi-hop cost-to-go)

delay, p99 $\sim$ BIND = $\max(PF, PL)$

nested search:
each PO $\uparrow$ own placement space